

Successfully transforming to Lean by changing the mindset in a global product development team

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Abstract—this paper describes the experience of a global software engineering organization in transforming to Lean. Deploying Lean processes did result in several improvements, but the primary goal—usable software at the end of each takt (iterations or increments, or single-piece flow)—was not achieved. We recognized that to achieve the primary goal and larger benefits of Lean, we needed to change the mindset of the global development team, and we defined approaches to enable this change. We have described the steps taken to make this change. The results achieved are also presented.

Keywords—Lean, global, software engineering, mindset, culture

I. INTRODUCTION

We are a global software development organization within a large multinational corporation that develops mission critical software where the cost of failure is very high. We had decided to transform to lean. Despite re-engineering the processes we still faced issues, and the main objective to have usable software after every takt was not achieved. In addition, several features were still spanning multiple takts. We realized that the essence of lean was not only the process but also the mindset—a mental model of how the world works [1].



Fig. 1. 80 percent of the Lean opportunity is in the culture [2]

In fact, it is estimated that effective enablement of Lean methodology depends 80 percent on culture and mindset, and only 20 percent on processes (see Fig. 1). We defined an approach to work on the changing the mindset.

II. BACKGROUND

The global development organization was distributed across centers in Asia and Europe. The delivery centers reported into the global headquarters (see fig. 2). Key experts in areas such as architecture, product management, usability, domain related, and system testing were at the headquarters.

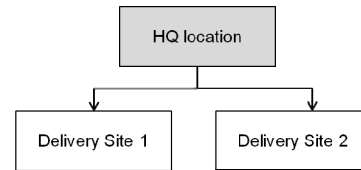


Fig. 2. The initial structure of the global development organization

Things were going well and multiple releases were successfully delivered. However, the development of an innovative new product saw the global development team's performance marred by significant time lags and cost overruns.

The organization quickly realized this was due to challenges introduced by the new product such as cyclic dependencies across the requirements of different sub-systems.

To correct this situation, it was decided to deploy Lean practices with emphasis on takts to ensure coordination across all locations. A three-month long pilot project using Lean was started and three takts were planned.

Based on the successful pilot, a complete project migrated to Lean. Subsequently, other projects followed. Within two years of deploying Lean we saw improvements in productivity, quality, and several other aspects [3].

However, we were still unable to have completely usable software at the end of each takt and had to introduce additional takts exclusively for testing and fixing defects.

III. CHALLENGES

Our approach for Lean is based on nine core principles [3]. We had deployed the Lean processes across the entire global team. An assessment of the degree of implementation of Lean processes, which examined aspects such as the process map, process implementation, and usage of templates, showed the processes were being followed.

This led us to realize there was a need to change the mindset of the global team so as to internalize six of the nine core elements (see Fig.3).

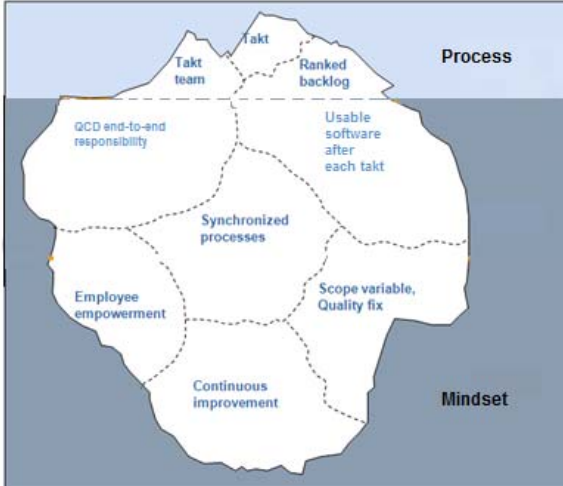


Fig. 3. Six of the nine core principles depended on mindset

It is known that people's beliefs have power and changing their beliefs—even the simplest beliefs—can have profound effects [4]. The teams worked on identifying their limiting beliefs [5] in each of the six mindset-dependent core principles of our approach. To do this, takt team speakers, Lean coaches, and seniors talked to the team members to identify what problems were limiting them with regard to the core principles.

A. Employee empowerment

In project terms we focused on four things. One, that we are one global team and equally responsible and accountable for the business results.

Two, every team member had to feel empowered to raise a red flag when a show-stopping problem was noticed. In the earlier operating model the red flags were only raised by managers and initially the team members were reluctant to do this as they considered raising a red flag as a responsibility of the managers. Overcoming this limiting belief required encouraging them to speak up when they saw problems in scope, quality issues, basically anything that could potentially lead to a project breakdown. In addition, it required creating a culture where team members felt free to ask questions.

Three, the management had to learn to accept the red flags raised for valid reasons, and support the team and avoid the

temptation to micromanage even when critical deadlines and milestones were delayed.

Four, takt team members had to work as a team and assume full responsibility for their takt results. They had to feel empowered to say “if we need to achieve this, this is what we need to do”. Previously, teams were more or less working in silos on tasks that were given by headquarters, and an “I did as was told to” way of working was common. To overcome the prevailing mindset, they had to feel empowered to work as a team to solve problems and stand up for decisions they took.

B. Continuous improvement

The aim was to make continuous improvement the core of our Lean journey and a part of our culture. In project terms, it meant creating more value through continuous improvement. It involved changing the mindset from local optimization to global systemic improvements and changing mindset of team members from “responsibility for a component” to “responsibility for the whole product”. This required team members to have a deeper understanding of the customer view of the requirements, and how these were met by the systems and solutions the team developed.

C. Quality is fixed (Scope variable, quality fix)

This core principle emphasizes that quality cannot be compromised. All committed work products (takt results) had to be delivered on time, within budget, with specified quality. In project terms, it meant that the tasks picked by the team members had to be completed with the required quality within the takt. In the past requirements often spanned multiple takts, which resulted in late integration of few features leading to quality issues. This change in mindset was hard for team members who now had to appreciate the urgency and need for intensive collaboration between all locations.

D. QCD (Quality, Cost, and Delivery):

The aim was to enable the mindset of all team members that they are fully accountable for the committed takt results delivered by them. This meant that team members had to identify the dependencies for their takt results and ensure that the committed features are delivered according to the quality, cost, and delivery criteria.

E. Synchronized Processes

The aim was to work hand in hand as one global team in a globally distributed organization. This meant the development of hardware, firmware, software, and end-user documentation had to be synchronized to ensure that at the end of each takt of equal duration, a usable version of the features was available. This required a mindset where all takt team synchronize their current status continually.

F. Usable software after each takt

Team members had to develop a mindset that at the end of each takt, the all committed takt results had to be integrated, system tested, and releasable to internal customer. One of the challenges was the ability to define small testable user stories,

which required breaking down larger functionality into smaller meaningful pieces across the entire product development.

IV. MEASURES

To overcome the limiting beliefs and changing the mindsets we defined a series of measures that were rolled out across all locations.

A. Measures to empower employees

1) Takt kick-off meeting:

In this meeting the commitment of the project takt team was communicated based on their assessment of feasibility. The project takt team had to take responsibility to deliver and honor the commitment with required quality. This enabled the mindset of one team with one goal.

2) Process to raise red flags:

This process helped team members to change their mindset and realize that raising a red to stop the line was everyone's responsibility. In addition, valid red flags raised were appreciated and supported by the key stakeholders such as R&D head and product manager.

3) Expert support:

This measure helped reinforce the mindset that team members were not alone in solving their day to day problems and had expert support. Team members were encouraged to reach out to expert in the communities of practice within the global team such as architecture, domain-specific topics, usability, while assessing feasibility and exploring solutions.

4) Competence development to take on global roles:

We augmented our organization-wide training program [6, 7], with the goal of providing team members cross-functional competences and behavioral skills to take on global roles. This helped develop mindset that we are one global team where all delivery centers have equal accountability and responsibility for business results (see fig 4). In addition, it helped creating mindset that global roles are open to any deserving candidate at any center

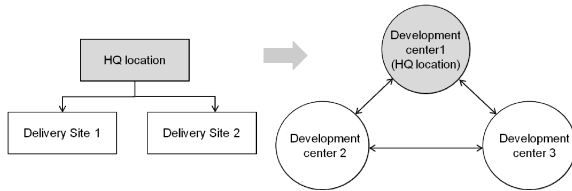


Fig. 4. Changing the mindset from delivery sites to one global team

The additional training focused on providing a deeper understanding of the complete system, the high-level architecture of the different components, detailed walkthrough of the designs of their specific components and modules. The training of team members was rigorously monitored by the project management team and was conducted by the experts in the team. The training also helped team members to effectively participate in different cross-functional teams.

5) Fun@work

Fun@work is part of the organization-wide 4 C approach [5]. The aim of fun@work was to enable the mindset that work is fun, creative and challenging-work is not monotonous. Previously team members considered work to be very task oriented and monotonous. In addition, the fun@work initiative helped team members in solving problems creatively. It also helped intensify communication, cooperation, and collaboration. Fun@work included a reward for the best team in a takt, badges signaling commitment to the takt goal, introducing quality mascots to communicate thoughts on quality, and identifying quality ambassadors to strengthen the culture of quality.

Empowerment stemmed from the new mindset of being one global team, the collaboration with team members within and across locations and the support that was global in nature. They knew they had the backing of their team while being responsible for their tasks.

B. Continuous improvement:

The goal was to develop a mindset that continuous improvement is everyone's responsibility and also nurture a sustainable culture across all levels through a structured approach. An outcome of this was a series of improvements which enabled the mindset of viewing problems through the eyes of customers. Previously all customer interactions and customer related information were only available at the headquarters. These include:

1) Defect analysis

Field defects were analyzed at a system level by a cross-functional team with representation from team members of all subsystems. In addition, the development team started to get monthly feedback from the Customer Care Department.

2) Job rotation

Project team members are deputed to the Customer Care Department for a few months to understand first-hand real customer issues and usage of the systems and solutions we develop.

3) Meeting with product management

Monthly meeting with the product management team were scheduled where the details of the market situation were shared and the development team could clarify requirements. This further strengthened customer orientation.

4) Participation in customer acceptance tests

Developers are sent to participate in customer acceptance tests so that they understand the real configurations which enabled them to improve many aspects of the software such as workflow management and testing.

5) First demo to product management team

Features were first demonstrated to product management and their feedback of the product management is incorporated in the next implementation of the feature.

These measures helped the team in viewing problem situations through the eyes of the customer. This helped the global team understand the ramifications of the problem and in proposing solutions which helped minimize local optimization by moving from “responsibility for component” to “responsibility for the whole product.”

C. *Quality is fixed:*

We introduced several measures to enable a mindset of quality within the team. This meant team members had to internalize that quality was priority. The committed quality for the takt results had to be achieved at the end of each takt. This meant that no open defects were permitted at release time and all requirements are successfully integrated. Teams had to assign the highest priority to open defects and re-rank new features.

1) *Quality gates*

The quality gate, a set of eleven quality-related key performance indicators (KPIs), was defined for the Takt goal. If the component did not achieve the goal it was not integrated. One of the KPIs was the number of defects

identified in system test. The goal for this KPI was zero severity 1 and severity 2 defects.

2) *Fixing defects*

Fixing defects was the highest priority in a takt and had to be done before implementing new features. Every team member had a quality, cost, deliverable (scope) target.

3) *Pick mandatory requirements to manage workload*

If the workload demanded, team members could only pick the mandatory tasks in a takt.

D. *QCD (Quality, Cost, Delivery)*

1) *Takt teams given end-to-end responsibility for features*

Every takt team was assigned end-to-end responsibility for customer features. The takt team had to take ownership of the features assigned and had to view the feature through the eyes of the end user. The team had to do what was necessary for the feature to do what was expected of product, which is the essence of end-to-end responsibility.

This measure helped move the team from “responsibility for component” to “responsibility for the whole product”.

TABLE I. SUMMARY OF CHALLENGES AND MEASURES

Core element	Aim	Challenge/Mindset change required	Measures/ Enablers
Employee empowerment	- Team members raise red flags when show-stopping problems are noticed. Management team accepts all red flags raised. - Takt team members had to work as a global team and assume full responsibility for their takt results.	- Overcoming team members' reluctance in taking on what was seen as a responsibility of a manager. - Management to accept the red flags raised for valid reasons, and support the team. - Moving away from the earlier “I did as I was told to” way of working.	- Takt kick-off meeting to communicate commitments. - Process to raise red flags introduced to encourage team members raising concerns as soon as they were detected. - Expert support to reinforce technical confidence in team members. - Intensified training to provide team members an understanding of the overall system and detailed insights in their specific components. - Fun@work initiatives to promote teamwork.
Continuous improvement	Value creation through continuous improvement.	- Make continuous improvement a part of our culture and this is responsibility of everyone in the organization. - Initial focus on viewing problems through the eyes of customers.	- Defect analysis by a cross-functional team with feedback from the Customer Care Department to understand the issues faced by customers. - Job rotation involving deputation to the Customer Care Department to sensitize the team members to end-user issues and the impact of defects. - Meeting with product management to share details on the market situation and clarify details. - Participation in customer acceptance tests to understand real configurations essential to improve workflows and testing. - First demo to product management team to incorporate feedback in the next takt.
Quality is fixed (Scope variable, Cost, quality fix)	All work products had to be delivered on time, within budget, with specified quality within the takt.	Quality cannot be compromised. Teams had to assign the highest priority to open defects and re-rank new features.	- Quality gates that ensure only code with acceptable quality is integrated. - Fixing defects prioritized ensures the focus is always on creating high-quality software at every takt. - Pick mandatory tasks only to manage workload.
QCD (Quality, Cost, Delivery)	Everyone is responsible for Quality, Cost, Delivery.	All team members feel that they are fully accountable for the committed takt results delivered by them.	Takt teams given end to end responsibility for features, which helped move the teams from component view to a product view.
Synchronized processes	All takt results to be synchronized across the global organization.	All takt teams synchronize their current status continually.	- Common processes, tools, KPIs, and takt duration. - Status calls involved all key stakeholders ensured synchronization at all levels. - Automation to enable continuous integration.
Usable software after each takt	At the end of each takt, software with committed features that met the quality criteria had to be delivered.	All committed takt results had to be integrated, system tested, and releasable to internal customer by the end of each takt.	- Formation of cross-functional takt teams instead of traditional sub-system specific takt teams. - Competence in writing smaller user stories that could be implemented, tested, and delivered in a single takt, by applying usage-centered design

E. Synchronized processes

We introduced measures to facilitate collaboration and cooperation between team members.

1) Common processes, tools, KPIs, and takt duration

All locations shared the same lifecycle processes, used the same tools and had the same KPIs. In addition, the takt duration was the same across all locations, which was sacrosanct.

2) Status calls involved all key stakeholders

Project status across all departments (engineering, product management, and customer care) was communicated at all levels through a global meeting attended by all key stakeholders ensured synchronization at all levels.

3) Automation

The teams practiced continuous integration which was enabled by automation of daily builds. The takt teams had to maximize the test automation to assess the quality of the feature developed. This included extensive unit test automation, automated smoke tests after build roll out, and integration test automation. Coupled with the defect prioritization, the test automation enabled the teams to complete the testing in every takt.

The synchronized processes help orchestrate the actions of the global team.

F. Usable software after each takt

1) Competence in writing smaller user stories

Competence in writing smaller testable user stories was important towards achieving this. A smaller user story is one that can be implemented, tested, and delivered with the required quality in one takt. Workshops were organized to train product owners, architects, domain experts, test experts, developers, usability engineers, user documentation experts in usage-centered design, a basis for smaller user stories [9].

2) Cross-functional takt teams

Over time the requirements were not only more cross-functional, but also introduced cyclic dependencies among the different sub-systems. To address this situation, cross-functional takt teams were formed. Previously, takt teams were formed around sub-systems because the sub-systems were largely developed in a single location. This change led to increased collaboration as the team was fully aware of the dependencies and could better plan their takt. This improvement also helped to overcome the limiting belief that multi-location takt teams cannot work.

V. RESULTS AND LESSONS LEARNED

The measures introduced enabled the global team in being more productive as is reflected in the results presented below.

A. Results

Mindset change works and is effective. The team delivered the higher quality, conforming to all release criteria. In one instance, the team delivered the product one week ahead of schedule, with 20 percent more requirements than committed at 10 percent less than the budgeted cost. The team engagement was very high reflected in a retention rate of over 95 percent.

B. Lessons learned

We present some of the things we would do if we had to re-live the journey.

1) Focus on mindset from the beginning

We started the migration to Lean with full focus on process but learned that, in a globally distributed development team it's more important to focus on changing the mindset.

2) Continuous improvement:

Continuous improvement is the core of the Lean journey.

3) Work as one team

In a globally distributed development organization, it's critical to work as one team.

4) Customer focus:

Always view problems through the eyes of customers

5) Fun@Work

Being people-centric business, it's essential that teams enjoy their work.

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